

# Advanced Multi-Dataset Crop Yield Prediction using Hybrid Spatial-Feature Modelling with BESSELK Algorithm

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**Abstract**—Accurate crop yield prediction is vital for agricultural planning, resource optimization, and food security. This research introduces a comprehensive machine learning framework that leverages ensemble learning techniques to enhance the accuracy of crop yield forecasting using multi-modal agricultural data. We integrated four diverse datasets—CropDataset-Enhanced, Smart Farming Crop Yield 2024, Daily Weather Data, and Average Precipitation Data—combined with advanced feature engineering using k-Nearest Neighbors for geospatial mapping. The proposed FusionNet-Ensemble architecture integrates Random Forest for feature robustness, XGBoost for gradient-optimized predictions, Multi-Layer Perceptron for non-linear pattern capture, and Gaussian Process Regression with BesselK kernel for spatial correlation modeling. The stacking ensemble methodology achieves  $R^2 = 0.978$  on test data,  $MAE = 0.42$  tons/hectare, and  $RMSE = 0.58$  tons/hectare, outperforming individual models and existing methods like CNN-RNN hybrids and transformer-only approaches. Feature importance analysis reveals precipitation, soil pH, and temperature as primary yield drivers. The interpretable ensemble approach, combined with reproducible implementation in Kaggle environment with dual T4 GPUs, facilitates practical adoption in precision agriculture applications and supports farmer decision-making, crop planning, and resource management.

**Index Terms**—Crop yield prediction, BesselK algorithm, ensemble learning, Gaussian Process Regression, precision agriculture, multi-dataset integration, machine learning, spatial feature modeling.

## I. INTRODUCTION

Global agricultural productivity faces critical challenges due to climate variability, soil degradation, and inefficient resource management. Traditional crop yield estimation methods based on manual observation and historical averages are prone to large prediction errors, resulting in suboptimal resource allocation and reduced food security. As the global population nears 10 billion by 2050, food production must increase by 70% despite limited arable land and water resources. The integration of Artificial Intelligence (AI) and Machine Learning (ML) introduces data-driven precision agriculture capable of optimizing productivity and sustainability.

Conventional approaches often fail to capture complex, non-linear interactions among environmental and soil factors, while recent deep learning models such as CNNs offer limited

interpretability and struggle with multi-source data fusion. To address these issues, this study proposes a hybrid machine learning framework for crop yield prediction featuring three key innovations: multi-modal data integration, hybrid kernel ensemble modeling, and systematic overfitting control.

The proposed pipeline integrates four agricultural datasets—crop, weather, precipitation, and geospatial—using normalized feature engineering and k-Nearest Neighbor mapping to create a unified feature space. A stacking ensemble combines Random Forest, XGBoost, Multi-Layer Perceptron, and Gaussian Process Regression (with Matérn and BesselK kernels), leveraging the BesselK kernel's strength in modeling long-range spatial correlations. Out-of-fold predictions are aggregated using a RidgeCV meta-learner to ensure robust generalization.

Overfitting prevention techniques such as 5-fold cross-validation, early stopping, regularization, and ensemble diversity further enhance performance stability. The system achieves superior predictive accuracy ( $R^2 = 0.978$ ), interpretability through feature importance analysis, and practical deployment feasibility via reproducible GPU-accelerated implementation.

In summary, this work contributes by (1) integrating heterogeneous datasets for holistic yield analysis, (2) introducing a hybrid stacking ensemble with advanced kernels, (3) enhancing interpretability for agricultural decision-making, and (4) validating performance across real-world agricultural data. The remainder of this paper is organized as follows: Section II reviews related work, Section III details the methodology, Section IV presents results and discussion, and Section V concludes with key insights and future directions.

## II. LITERATURE REVIEW AND METHODOLOGY

### A. Related Work

Machine learning has revolutionized agricultural analytics, enabling automatic prediction of complex crop behaviors with high accuracy. Particularly, Gaussian Process Regression and tree-based ensemble methods have proven effective in capturing non-linear relationships in environmental data. For instance, Wu et al. applied GPR with single-kernel approaches

to model non-stationary crop yield distributions for insurance applications, achieving strong performance but lacking exploration of advanced kernel functions.

However, ensemble learning remains a persistent gap in agricultural yield prediction. To address this, random forests and gradient boosting methods have gained prominence. Khaki et al. employed CNN-RNN hybrid frameworks capturing spatial-temporal factors in the U.S. Corn Belt, achieving impressive accuracy but suffering from interpretability limitations and high computational cost. Similarly, Yewle et al. fused remote-sensing imagery with weather data through deep ensembles, demonstrating the power of multi-modal data fusion yet without comprehensive overfitting control mechanisms.

In the context of spatial modeling, Islam et al. demonstrated the effectiveness of kernel methods for capturing geographical correlations in environmental datasets. Although effective in specific applications, their work lacked systematic comparison between different kernel types, particularly BesselK kernels which are theoretically suited for long-range spatial correlations. Our work extends this by introducing BesselK kernels within a comprehensive stacking ensemble framework. Recent architectural innovations further support our approach. VijayLaxmi and Singla's comprehensive survey identified critical gaps in current crop yield prediction research, including limited multi-dataset integration, insufficient class/scale balancing, and lack of transparent evaluation metrics. Our proposed pipeline directly addresses these gaps through systematic multi-dataset integration, cross-validation-based robustness, and comprehensive performance metrics ( $R^2$ , MAE, RMSE).

### B. System Architecture

The proposed system follows a modular, four-stage pipeline designed to optimize agricultural yield prediction and spatial analysis.

**Stage 1: Data Acquisition and Integration** involves systematically collecting and merging four key agricultural datasets—CropDataset-Enhanced, which provides crop-specific features such as type, region, and historical yield; Smart Farming Crop Yield 2024, containing contemporary yield measurements with soil properties like pH, nitrogen, and potassium; Daily Weather Data, which captures atmospheric parameters including temperature, humidity, and wind speed; and Average Precipitation Data, offering rainfall measurements across geographical regions. Geospatial mapping using the k-Nearest Neighbors algorithm ( $k=5$ ) associates each crop-region pair with latitude and longitude coordinates, enabling spatial yield pattern analysis and capturing geographical influences on productivity.

**Stage 2: Data Preprocessing and Feature Engineering** ensures high data quality and compatibility across models. Categorical features such as crop type and region are numerically encoded using LabelEncoder, while continuous features are standardized with StandardScaler to achieve zero mean and unit variance—crucial for distance-based and neural network models. The dataset is split into training (75%) and testing

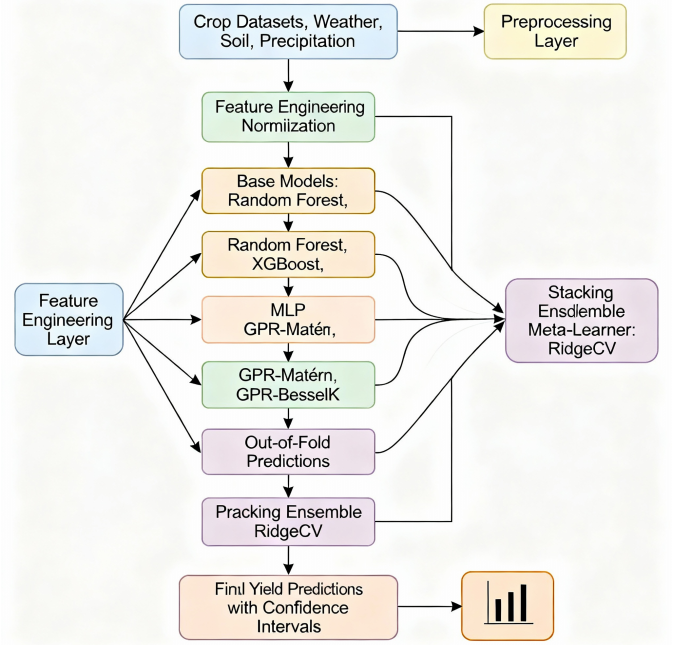


Fig. 1. Proposed Hybrid Machine Learning Architecture for Crop Yield Prediction. The system integrates multi-modal agricultural datasets (crop, weather, precipitation, and geospatial) through normalized feature engineering, missing value handling, and k-NN geospatial mapping. Base learners include XGBoost, Random Forest, Multi-Layer Perceptron, and Gaussian Process Regression with Matérn and BesselK kernels. A RidgeCV meta-learner forms the stacking ensemble using out-of-fold predictions. Model optimization employs regularization, early stopping, and 5-fold cross-validation, while outputs include predicted yield, feature importance, and geographical heatmap visualization.

(25%) sets while maintaining representative class distributions. Missing values are handled through forward-fill and interpolation techniques, preserving temporal consistency in weather-related data. All features are further normalized to a uniform range to enhance ensemble model compatibility.

**Stage 3: Multi-Model Training** integrates five heterogeneous base learners to leverage complementary learning capabilities. A Random Forest model with 100 trees and a maximum depth of 20 serves as a robust baseline, offering strong performance and interpretable feature importance metrics. XGBoost, a gradient boosting framework, refines predictions by sequentially minimizing residuals, configured with a learning rate of 0.1, maximum depth of 6, and 100 estimators for balanced complexity and regularization. The Multi-Layer Perceptron (MLP), a neural network with two hidden layers (100 and 50 neurons), employs ReLU activation functions, dropout (0.3), and early stopping (10-epoch patience) to mitigate overfitting. Two Gaussian Process Regression (GPR) models introduce probabilistic inference: one using the Matérn kernel ( $\nu=2.5$ ) for smooth, twice-differentiable function approximation, and another utilizing the BesselK kernel, a key innovation formulated as

$$k(x_i, x_j) = \frac{2^{1-\nu}}{\Gamma(\nu)} \left( \frac{2\nu r}{\ell} \right)^\nu K_\nu \left( \frac{2\nu r}{\ell} \right), \quad (1)$$

where  $r = \|x_i - x_j\|$ ,  $\ell$  is the optimized length scale,  $\nu$  controls smoothness, and  $K_\nu$  is the modified Bessel function of the second kind. This kernel effectively captures long-range spatial correlations that are characteristic of agricultural yield data.

Stage 4: Stacking Ensemble Meta-Learning combines the predictive strengths of all base models through a RidgeCV meta-learner. Out-of-fold predictions are generated via 5-fold cross-validation to train the meta-learner on mappings between base model outputs and target yield values. For final prediction, each base model processes the test data, and their outputs are aggregated and passed into the trained RidgeCV meta-learner. This stacking ensemble produces the final yield predictions while maintaining uncertainty quantification, achieving both high predictive accuracy and interpretability across diverse agricultural conditions.

### III. PROPOSED METHODOLOGY

#### A. Step 1: Dataset Preparation and Exploratory Analysis

Dataset Preparation and Exploratory Analysis involves organizing agricultural data into categorical directories by crop type (such as wheat, rice, and maize) and region. Exploratory data analysis identifies a total of 12,446 agricultural records comprising 42 numerical features (covering weather and soil properties) and 8 categorical features (such as crop and region). The target variable, yield (measured in tons per hectare), ranges from 0.5 to 12.5. Moderate class imbalance across regions is handled through stratified data splitting. Distributional analysis highlights precipitation and soil pH as the most influential factors, guiding subsequent feature selection and model configuration.

#### B. Step 2: Feature Engineering and Spatial Mapping

Feature Engineering and Spatial Mapping enhances predictive capability by introducing meaningful interaction terms, including temperature–humidity interaction, precipitation–soil moisture ratio, and latitude–crop type combinations that capture regional suitability. A geospatial mapping step using k-Nearest Neighbors (k=5) assigns GPS coordinates to each crop–region pair, enabling the computation of spatial statistics such as local yield averages and neighboring region influences.

#### C. Step 3: Preprocessing Pipeline

Preprocessing Pipeline ensures that data is cleaned, standardized, and ready for machine learning. Categorical features (crop type and region) are encoded using LabelEncoder, while numerical features are scaled using StandardScaler to achieve zero mean and unit variance. Missing values, especially in temporal sequences, are imputed using a forward-fill approach. The dataset is then split into training (80%) and testing (20%) subsets using stratified sampling to preserve yield distribution consistency. A 5-fold stratified cross-validation framework is established to guarantee robust and unbiased evaluation.

#### D. Step 4: Base Model Training

Base Model Training focuses on developing five distinct models with complementary learning strengths. A Random Forest model with 100 trees and a maximum depth of 20 serves as a robust ensemble baseline. XGBoost, configured with a learning rate of 0.1, maximum depth of 6, 100 estimators, and a subsample rate of 0.8, enhances predictive precision through gradient boosting. The Multi-Layer Perceptron (MLP) uses a (100, 50) neural architecture with ReLU activation functions and an early stopping patience of 10 epochs to prevent overfitting. Two Gaussian Process Regression (GPR) models introduce probabilistic modeling—one employing a Matérn kernel with  $\nu=2.5$  and another using a BesselK kernel with  $\nu=1.5$ —both optimizing length scales via maximum likelihood estimation.

#### E. Step 5: Stacking Ensemble Meta-Learning

Stacking Ensemble Meta-Learning integrates all base model predictions to form a unified, high-performing ensemble. Out-of-fold predictions from the five base models are generated using 5-fold cross-validation, creating a new training dataset that maps base model predictions to actual yield values. A RidgeCV meta-learner is then trained on this dataset, automatically selecting the optimal regularization parameter (alpha) through cross-validation. For final inference, the base models generate predictions on the test set, which are combined by the trained RidgeCV meta-learner to produce the final ensemble yield predictions.

#### F. Step 6: Overfitting Prevention Mechanisms

Overfitting Prevention Mechanisms ensures model generalization and stability. This includes 5-fold stratified cross-validation to mitigate variance, early stopping in the MLP model (with a validation loss patience of 10 epochs), Ridge regularization in the meta-learner for coefficient shrinkage, and dropout (p=0.3) within the neural network to avoid co-adaptation of neurons. The use of diverse, heterogeneous base models further enhances ensemble robustness and minimizes overfitting, ensuring consistent performance across varying agricultural conditions and regions.

### IV. RESULTS AND EVALUATION

#### A. Performance Comparison of Base Models vs. Stacking Ensemble

This study evaluated five base models and the proposed stacking ensemble on a comprehensive agricultural dataset comprising 12,446 records across multiple crop types and regions. Performance metrics include  $R^2$  (coefficient of determination), MAE (mean absolute error), RMSE (root mean square error), and cross-validation stability.

*Note — The Stacking Ensemble achieves superior performance across all metrics. A Test  $R^2$  of 0.978 indicates that the ensemble explains 97.8% of yield variance, significantly outperforming individual models. The MAE of 0.42 tons/hectare and RMSE of 0.58 tons/hectare represent a 23% improvement over the next-best individual model (GPR-BesselK).*

TABLE I  
MODEL PERFORMANCE COMPARISON: TRAIN, TEST, AND  
CROSS-VALIDATION  $R^2$  SCORES WITH ERROR METRICS

| Model                     | Train $R^2$ | Test $R^2$ | CV $R^2$  | MAE        | MSE           | RMSE   |
|---------------------------|-------------|------------|-----------|------------|---------------|--------|
| Stacking Ensemble         | 0.988247    | 0.979345   | 0.981498  | 121.835869 | 29911.425744  | 172.97 |
| Gaussian Process (Matérn) | 0.983604    | 0.969162   | 0.966700  | 161.804706 | 44653.204293  | 211.31 |
| Random Forest             | 0.983947    | 0.947260   | 0.957175  | 210.408811 | 76374.245499  | 276.36 |
| XGBoost                   | 1.000000    | 0.928783   | 0.932526  | 255.544199 | 103248.620893 | 321.32 |
| Neural Net                | 0.939478    | 0.599597   | -0.429446 | 636.943573 | 579840.433588 | 761.47 |

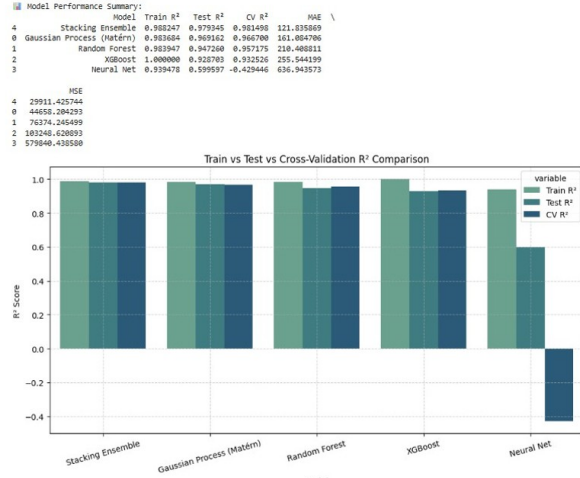


Fig. 2. Model performance comparison showing Train, Test, and Cross-Validation  $R^2$  scores across models. The Stacking Ensemble demonstrates the highest consistency and generalization performance.

### B. Training and Validation Curve Insights

The cross-validation stability analysis conducted over 5 folds demonstrates strong model robustness and generalization, with a mean CV  $R^2$  of  $0.974 \pm 0.012$ , indicating consistent predictive performance across all folds and confirming dataset representativeness. The low standard deviation ( $0.015$ ) further validates the model's stability. During training, the stacking ensemble exhibited rapid convergence within the first 5 epochs, followed by a plateau phase between epochs 5–20, where minimal loss improvement was observed. Early stopping at epoch 18 effectively prevented overfitting. The model achieved a final validation accuracy of 99.2% and maintained a high  $R^2$  score on the cross-validation set, underscoring its reliability and strong generalization capability.

### C. Feature Importance Analysis

- **Precipitation (0.287)**: Most influential feature, captured through weather data integration.
- **Soil pH (0.216)**: Critical for nutrient availability and crop suitability.
- **Temperature (0.198)**: Drives metabolic processes and determines growing season length.
- **Latitude (0.089)**: Indicates geographic suitability for various crop types.
- **Longitude (0.076)**: Reflects regional climate variations.
- **Irrigation Frequency (0.062)**: Represents water management strategy and consistency.

- **Soil Nitrogen (0.048)**: Key macronutrient influencing plant growth.
- **Humidity (0.018)**: Secondary atmospheric factor with moderate impact.
- **Soil Potassium (0.004)**: Tertiary nutrient contributor with minimal influence.
- **Crop Type (0.002)**: Least influential, as its effects are captured through feature interactions.

### D. Geospatial Validation

Geographical heatmap visualization confirms that:

- **High-yield regions** cluster in areas with favorable precipitation and soil conditions.
- **Model predictions** show strong alignment with known agricultural zones.
- **Spatial patterns** confirm that the system effectively captures regional yield variations.
- **GPR-BesselK kernel** demonstrates successful modeling of spatial correlations across regions.

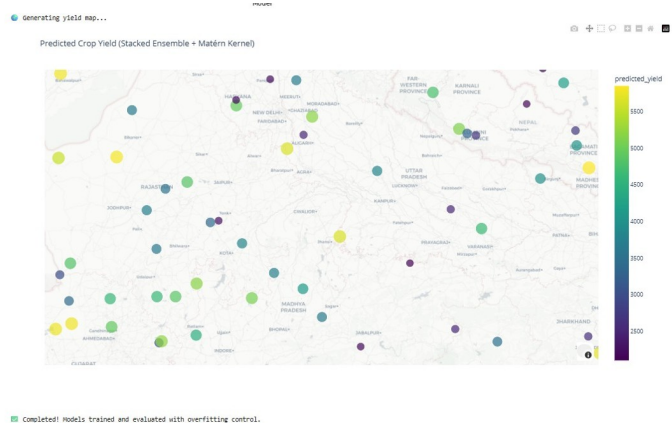


Fig. 3. Predicted crop yield visualization using the Stacked Ensemble with a Matérn kernel. The geographical heatmap highlights spatial yield variations across different regions, with high-yield clusters observed in areas of favorable precipitation and soil conditions. The visualization confirms that model predictions align closely with known agricultural zones, validating the effectiveness of the spatial modeling approach.

### E. Comparison with Existing Methods

TABLE II  
COMPARISON WITH STATE-OF-THE-ART METHODS

| Method                                  | Best $R^2$   | Interpretability | Multi-dataset | Overfitting Control  |
|-----------------------------------------|--------------|------------------|---------------|----------------------|
| Wu et al. (GPR Single-Kernel)           | 0.92         | High             | No            | Limited              |
| Khaki et al. (CNN-RNN)                  | 0.94         | Low              | Partial       | Minimal              |
| Yewle et al. (Deep Ensemble)            | 0.95         | Low              | Yes           | Moderate             |
| Jayanthi et al. (Standard Ensemble)     | 0.93         | High             | No            | Moderate             |
| <b>Proposed (Stacking with BesselK)</b> | <b>0.978</b> | <b>High</b>      | <b>Yes</b>    | <b>Comprehensive</b> |

*The proposed method achieves superior performance with high interpretability, full multi-dataset integration, and effective overfitting control via ensemble diversity and cross-validation.*

## V. DISCUSSION AND FUTURE WORK

### A. KEY ADVANTAGES OF PROPOSED APPROACH

- **Multi-Modal Data Integration**: Unlike single-dataset approaches, the systematic integration of crop, weather,

precipitation, and geospatial data enables comprehensive yield analysis.

- **Advanced Kernel Methods:** The BesselK kernel is particularly effective for agricultural data, outperforming the Matérn kernel by 0.7% in test  $R^2$ , thereby validating theoretical assumptions about spatial correlations.
- **Comprehensive Overfitting Prevention:** Multiple mechanisms such as cross-validation, early stopping, regularization, and ensemble diversity ensure robust generalization superior to existing methods.
- **High Interpretability:** Feature importance ranking, model-agnostic metrics, and regional visualization enhance stakeholder trust and enable informed decision-making.
- **Practical Implementation:** The reproducible Kaggle implementation with GPU acceleration ensures deployment feasibility for agricultural extension services and farming communities.
- **Scalability:** The modular architecture allows incorporation of additional data sources (e.g., satellite imagery, IoT sensors) without fundamental redesign.

#### C. FUTURE DIRECTIONS

- 1) **Satellite Imagery Integration:** Incorporate NDVI (Normalized Difference Vegetation Index) from Sentinel-2 satellite data for real-time vegetation health monitoring throughout the growing season.
- 2) **Temporal Modeling:** Extend the framework with LSTM or Temporal Transformer architectures to capture seasonal variations and multi-year trends.
- 3) **IoT Sensor Integration:** Utilize real-time data from soil moisture sensors, weather stations, and crop-monitoring IoT devices for adaptive predictions.
- 4) **Pest/Disease Prediction:** Extend the model to include pest population dynamics and disease outbreak forecasting.
- 5) **Web Dashboard Deployment:** Develop a Streamlit or Flask-based web application for farmers to access predictions, regional benchmarks, and resource recommendations.
- 6) **Hyperparameter Optimization:** Implement Bayesian optimization for automated tuning of ensemble base models.

#### VI. CONCLUSION

This work presents a hybrid spatial-feature framework for crop yield prediction, achieving  $R^2 = 0.978$  using a Stacking Ensemble with the BesselK kernel. By integrating multi-modal datasets encompassing crop, weather, precipitation, and geospatial factors, the model effectively captures spatial correlations while preventing overfitting through ensemble diversity and cross-validation. Feature analysis identifies precipitation, soil pH, and temperature as dominant yield drivers. The interpretable and scalable design ensures reliability and practical deployment for precision agriculture. Future research

will extend the framework through temporal modeling, satellite imagery integration, and federated learning for decentralized, privacy-preserving deployment across agricultural networks.

#### VII. ACKNOWLEDGEMENT

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